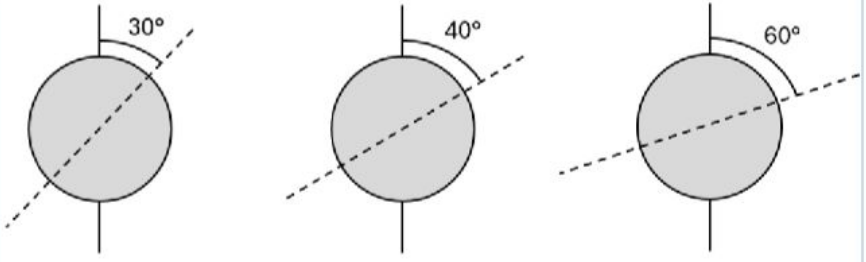
$E_{total} = \frac{1}{2}mv_0^2 = \frac{1}{2}m(\omega x_0)^2 \propto \left(\frac{x_0}{T}\right)^2 \Rightarrow \frac{E_{new}}{E_{old}} = \left(\frac{1}{3}\right)^2 = \frac{1}{9} \Rightarrow E_{new} = \frac{1}{9}E_{old} = \frac{1}{9}(18\text{ mJ}) = 8\text{ mJ}$

11 The diagram below shows three polarisers W, X and Y.

A beam of unpolarised light of intensity I_0 is incident normally onto the surface of polariser W, which then passes through polariser X and eventually emerges from polariser Y.

The axis of polarisation of each polariser is indicated by a dashed line.



Ans: C
Angle between successive polarisers:

- W to X: 40° – 30° = 10°
- X to Y: 60° – 40° = 20°

After Polariser W
Unpolarised light passing through the first polariser loses half its intensity: $I_W = \frac{I_0}{2} = 0.5I_0$

After Polariser X (Malus's Law)

$I_X = I_W \cos^2(10°)$
$= 0.5I_0 \times \cos^2(10°)$
$= 0.5I_0 \times 0.9698$
$= 0.4849I_0$

After Polariser Y (Malus's Law)

$I_Y = I_X \cos^2(20°)$
$= 0.484I_0 \times \cos^2(20°)$
$= 0.484I_0 \times 0.8830$
$= 0.428I_0 \approx 0.43I_0$

What is the intensity of the light that emerges from polariser Y?

A $0.11I_0$ **B** $0.32I_0$ **C** $0.43I_0$ **D** $0.64I_0$